

Crime Patterns Across Atlantic Canada: Clustering and PCA Analysis (2020–2024)

This document presents a clustering and PCA analysis of crime data in Atlantic Canada (2020–2024). It examines absolute crime levels, offence composition, and growth patterns to highlight trends and high-crime localities. Silhouette scores and PCA scatter plots visualize the cluster structures across regions.

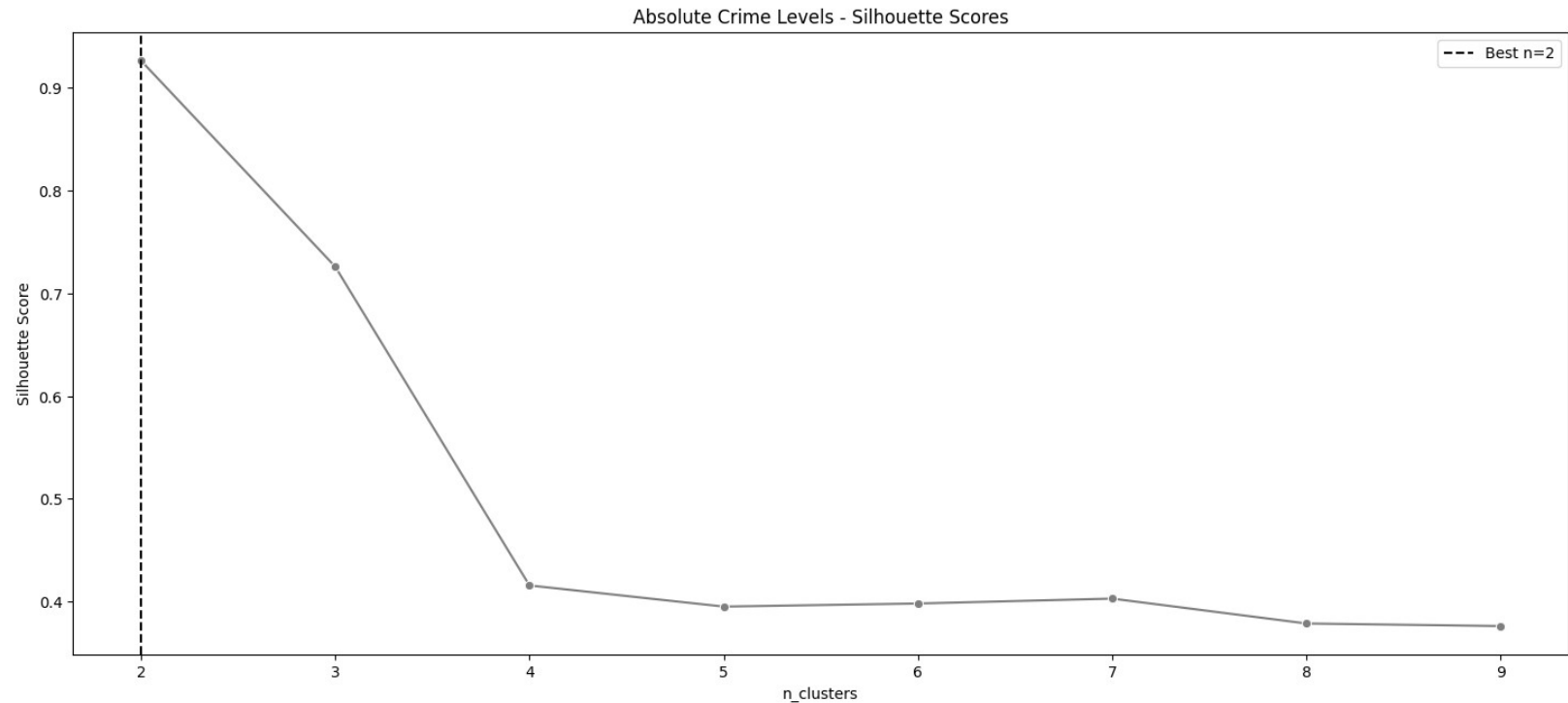


Figure 1

Silhouette Scores – Absolute Crime Levels

Silhouette scores for clustering localities based on absolute crime levels (2020–2024). The plot shows the optimal number of clusters for identifying low- and high-crime areas.

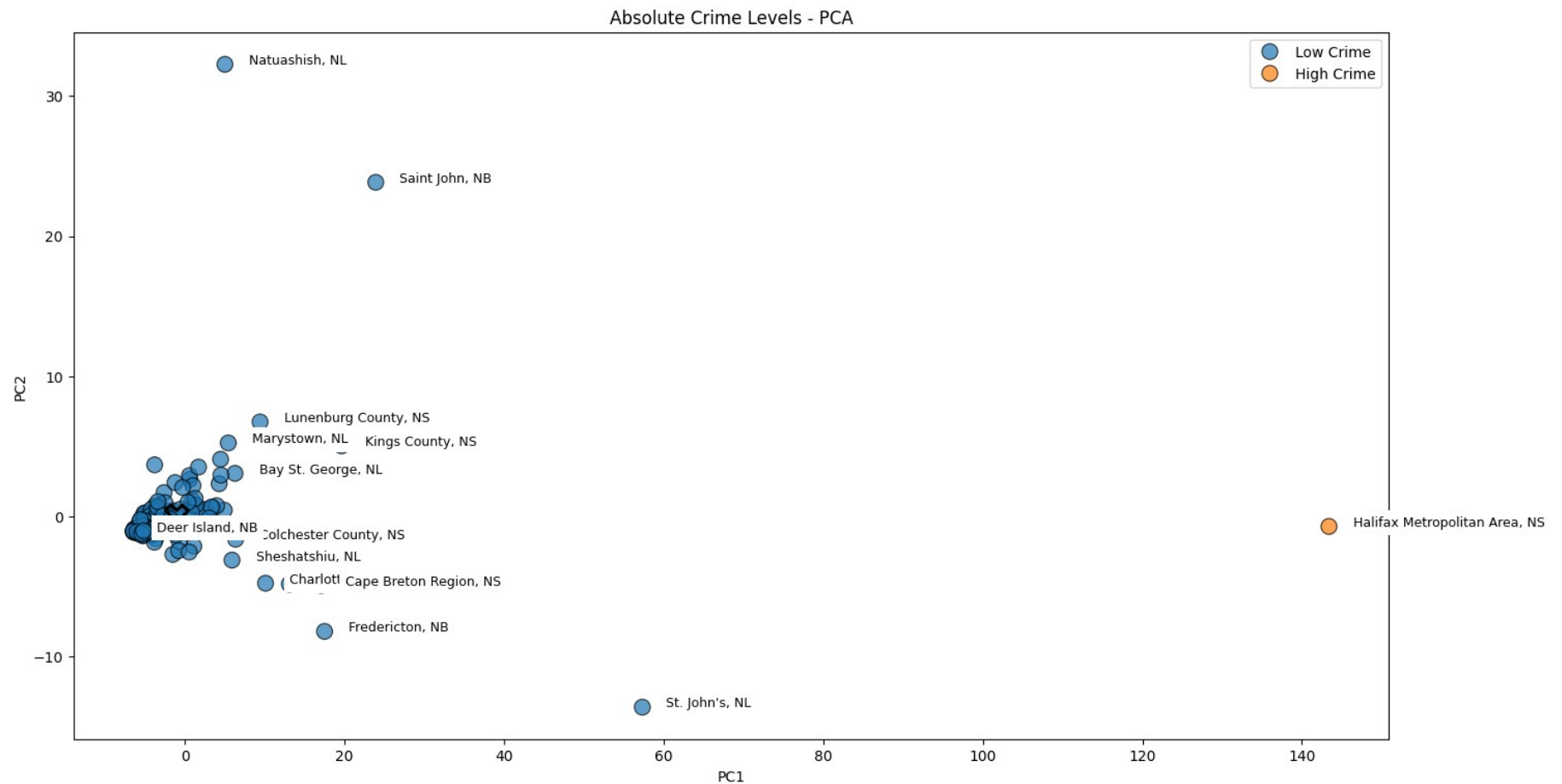


Figure 2

PCA Scatter – Absolute Crime Levels

PCA scatter plot of localities based on absolute crime levels. Points are colored by cluster, and high-crime localities are labeled to highlight areas with elevated criminal activity.

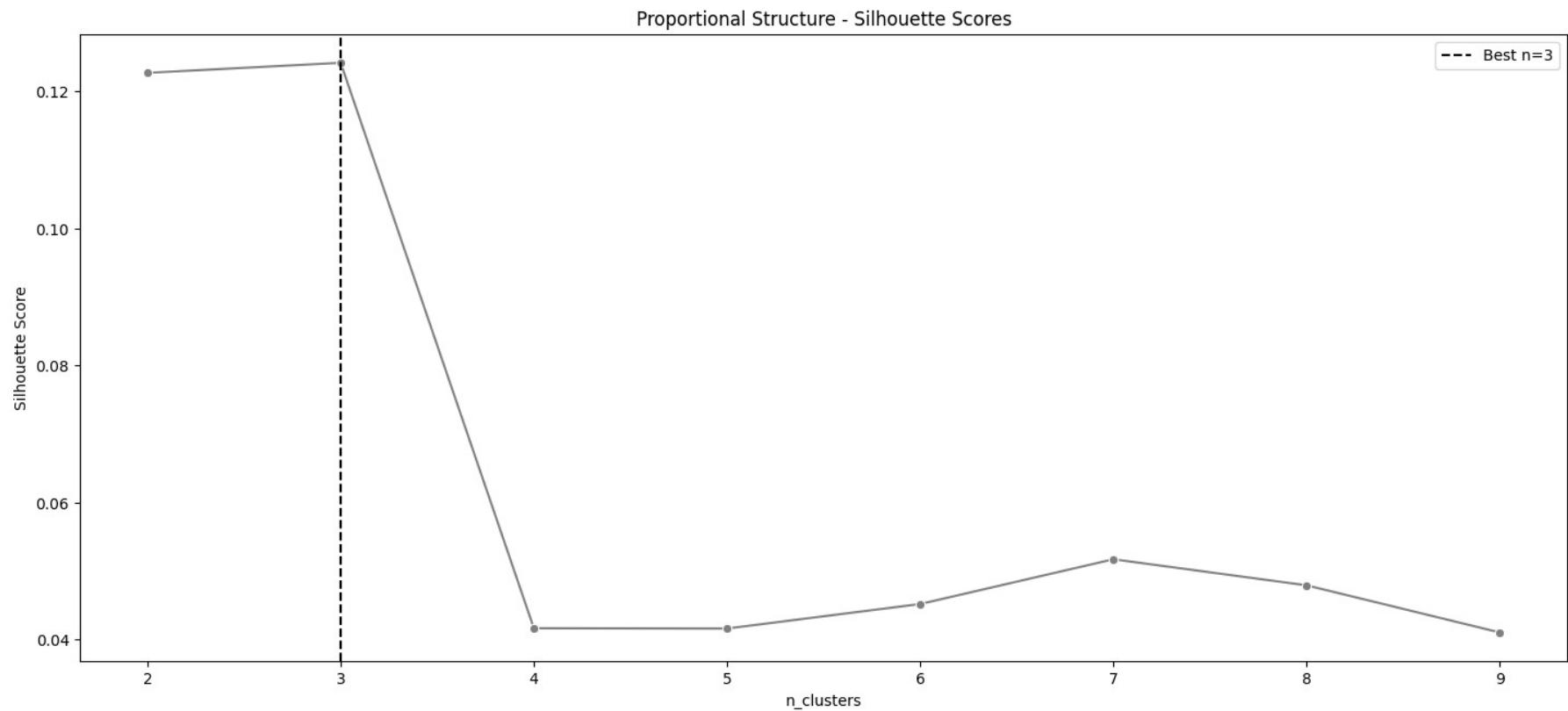


Figure 3

Silhouette Scores – Proportional Structure

Silhouette scores for clustering localities based on the proportional structure of offences. The graph indicates the best cluster configuration for comparing crime composition across regions.

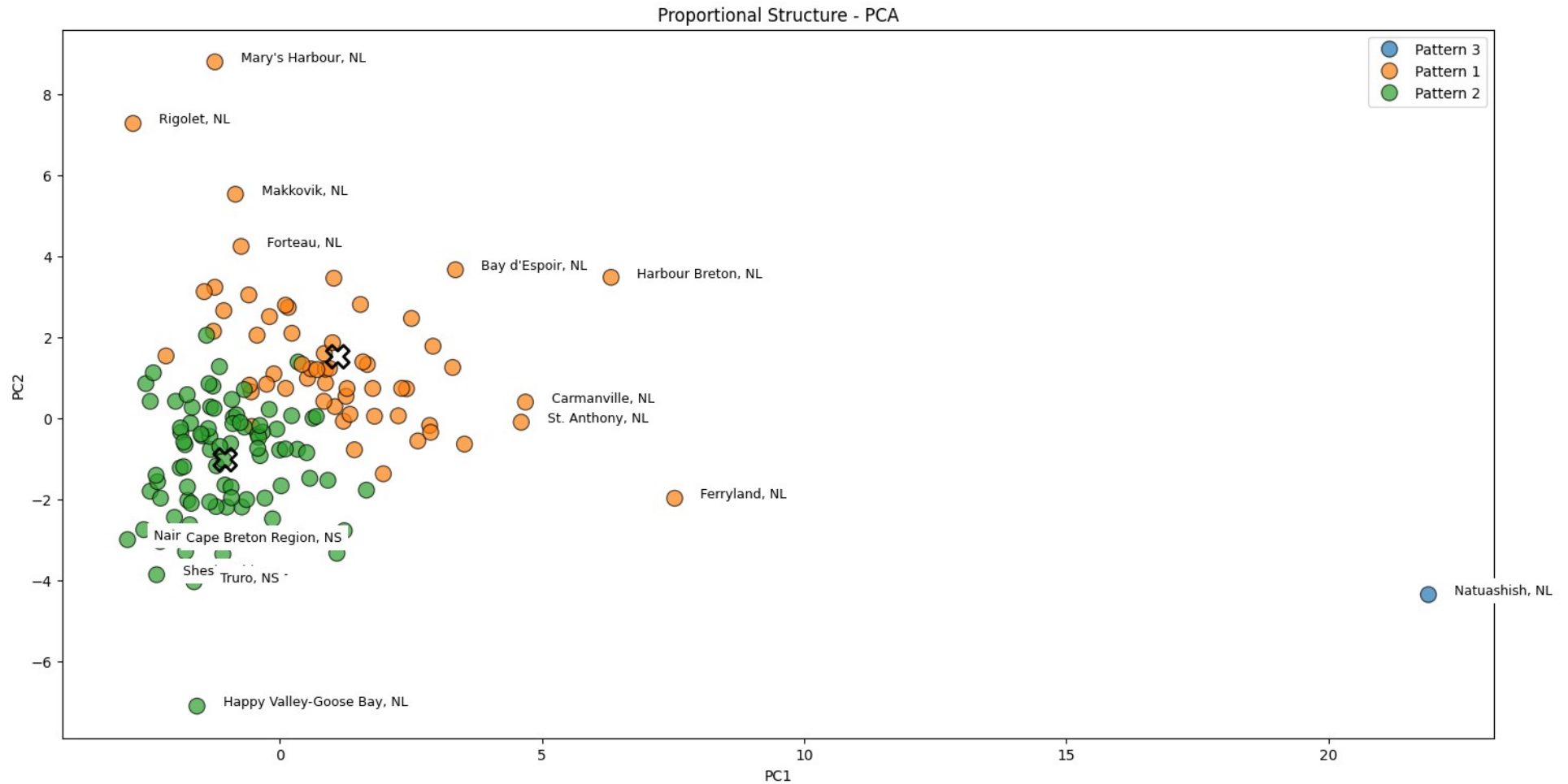


Figure 4

PCA Scatter – Proportional Structure

PCA scatter plot of localities based on the relative composition of offences. Clusters show groups of regions with similar crime patterns, with annotations for notable high-crime localities.

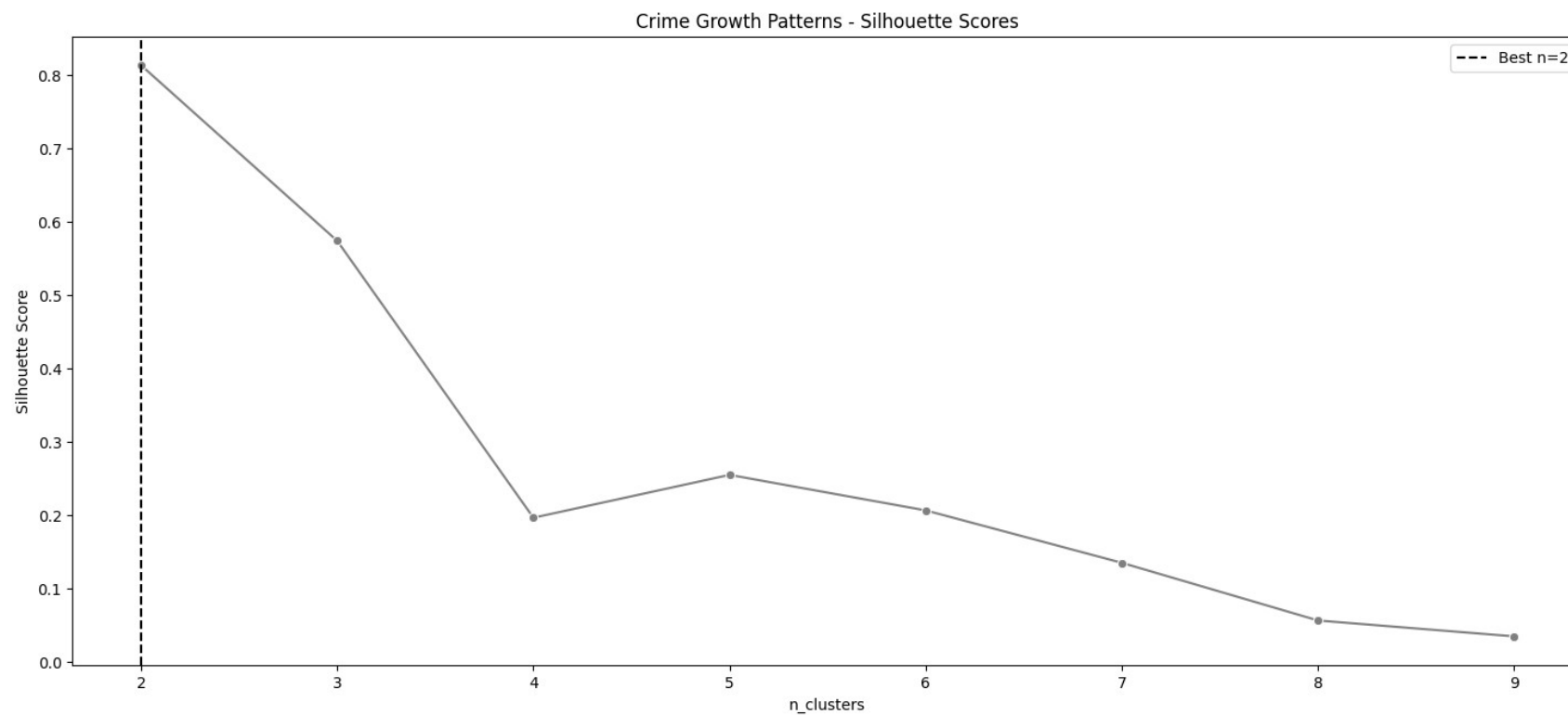


Figure 5

Silhouette Scores – Crime Growth Patterns

Silhouette scores for clustering localities according to crime growth trends over 2020–2024. The plot identifies the optimal number of clusters to distinguish areas with increasing or decreasing crime rates.

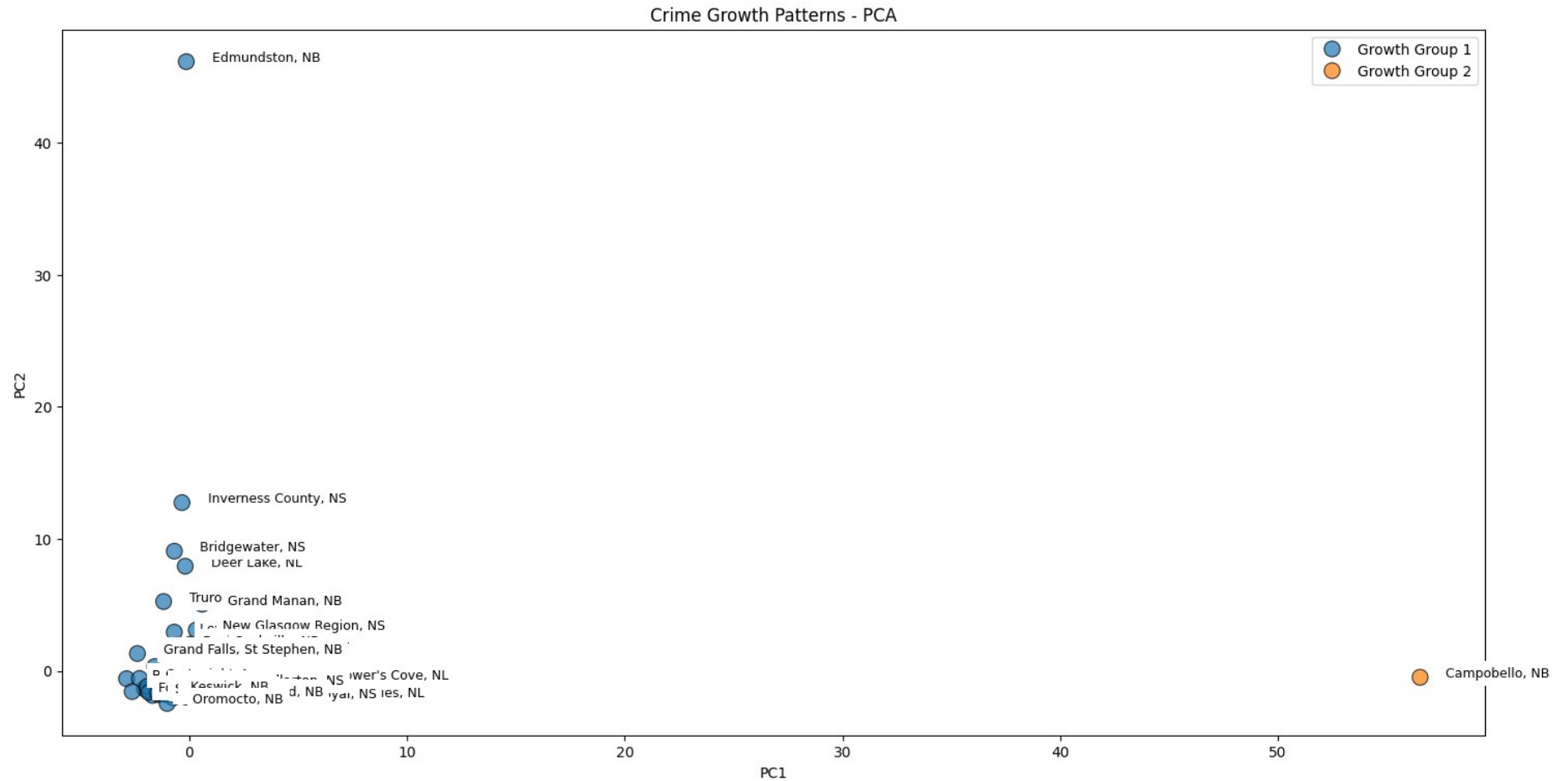


Figure 6

PCA Scatter – Crime Growth Patterns

PCA scatter plot of localities based on crime growth patterns. Points are colored by cluster, and localities with significant growth or reduction in crime are labeled for reference.